

June 10, 2026

Editor-in-Chief
Applied Sciences
MDPI

Dear Editor,

I am pleased to submit our manuscript entitled “Deep Learning for Student Engagement Detection: A Lightweight CNN Framework with Channel-Spatial Attention for Educational Environments” for consideration for publication in *Applied Sciences*.

This paper proposes a lightweight deep learning framework integrating MobileNetV3 with a novel Channel-Spatial Attention Module (CSAM) for real-time student engagement recognition from facial expressions. The main contributions are:

1. A lightweight CSAM module designed for mobile architectures, achieving 1.3–5.0% accuracy improvement with only 0.01M additional parameters.
2. Comprehensive evaluation across three datasets: FER2013 (70.5% accuracy), CK+ cross-dataset (71.9%), and DAiSEE (61.2% binary).
3. Deployment-oriented evaluation including parameter count, FLOPs, inference latency, and energy consumption.

The MobileNetV3+CSAM model achieves competitive accuracy with only 1.26M parameters and 53 fps, making it suitable for edge deployment. Both authors have approved the manuscript and declare no conflicts of interest.

I look forward to your decision.

Sincerely,

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